ELKANAH NYABUTO

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RESEARCH PROFILE

PhD candidate in Statistics at the University of Glasgow (thesis submitted, viva pending), specialising in spatial and spatiotemporal statistics, statistical learning, and data-driven estimation of complex dependence structures. My research develops regularised methods for learning spatial and network dependence directly from data, with applications in environmental science, financial networks, and infectious-disease epidemiology. This work has resulted in a peer-reviewed publication and ongoing manuscripts and has involved methodological development, computational implementation, Monte Carlo validation, empirical analysis, scientific writing, and international dissemination. I am proficient in R and Python, with additional experience in interdisciplinary research, university teaching, research capacity building, and scientific communication.

RESEARCH INTERESTS

Spatial & Spatiotemporal Statistics | Statistical Learning & Regularisation | High-Dimensional Statistics | Spatial Dependence and Network Learning | Time-Series Modelling | Computational Statistics | Environmental Statistics & Epidemiological Modelling

EDUCATION

University of Glasgow, Glasgow, UK
PhD in Statistics (Thesis submitted; viva pending) *2023–2026*
Thesis: *Spatiotemporal Dependence Modelling; Regularised Estimation of Spatial Dependence Matrices.*
Initially registered as a doctoral researcher at Leibniz University Hannover, Germany (2022–2023), before continuing the PhD at the University of Glasgow following the transfer of the research project.

African Institute for Mathematical Sciences, AIMS Ghana, Accra, Ghana
MSc. Mathematical Sciences *Jul 2020*
Thesis: Estimation of COVID-19 Patient Survival Times Using Kaplan-Meier Methods: Wuhan Case Study

Maseno University, Kisumu, Kenya
BSc Applied Statistics (Minor: IT) *Dec 2018*

HONOURS & AWARDS

DFG Scholarship – Statistical Learning of High-Dimensional Spatial Dependence Structures
University of Glasgow, UK 2023–2025
Two-year doctoral scholarship funded through a German Research Foundation (DFG) research project, covering a maintenance stipend and tuition fees.

Mastercard Foundation Scholarship 2019–2020
African Institute for Mathematical Sciences (AIMS), Ghana
Fully funded scholarship for postgraduate study in Mathematical Sciences, awarded with an emphasis on academic excellence, leadership potential, community service and commitment to Africa's development.

RESEARCH EXPERIENCE

University of Glasgow, Glasgow, UK Oct 2023–Present
PhD Researcher, School of Mathematics & Statistics

- Develop statistical methodology for learning unknown spatial and spatiotemporal dependence structures using regularisation and sparse estimation.
- Conduct methodological development, Monte Carlo simulation, numerical optimisation, predictive validation, and empirical analysis in R and Python.
- Apply developed methods across environmental statistics, financial networks, and infectious-disease epidemiology, resulting in peer-reviewed and ongoing research outputs.

University of Glasgow, Glasgow, UK *Sep 2025 – Jan 2026*

Research Assistant (Part-time) , School of Mathematics and Statistics

- Developed regularised statistical methodology for learning unknown spatial and spatiotemporal dependence structures in high-dimensional models using penalised likelihood and quasi-likelihood estimation.
- Designed Monte Carlo experiments evaluating parameter estimation, network recovery, predictive performance, stability and computational efficiency.
- Implemented numerical optimisation, temporal cross-validation, forecasting and model-selection procedures in R and Python.
- Applied and validated the methodology using environmental and financial data, comparing learned dependence structures with predefined alternatives through out-of-sample forecasting and statistical tests.

Leibniz University Hannover, Hannover, Germany *Sep 2022 – Sep 2023*
Research Assistant – DFG-funded project, Institute of Cartography and Geoinformatics

- Conducted statistical analysis of high-dimensional environmental datasets using spatial and spatiotemporal statistical models.
- Investigated data-driven estimation of spatial and temporal dependence using regularisation methods.
- Developed reproducible statistical workflows in R for data processing, model estimation, validation, and visualisation.
- Collaborated within an international research environment spanning spatial statistics, environmental modelling, and computational methods.

Maseno University Lab for Interdisciplinary Statistical Activities (MU-LISA) *Oct 2020 – Nov 2021*
Research Analyst & Statistician

- Contributed to multidisciplinary research through development of research proposals, study protocols, statistical analysis plans and grant applications.
- Conducted statistical analysis, interpretation and reporting in R, Python and SPSS, developing reproducible analytical workflows and research visualisations.
- Mentored undergraduate interns and delivered capacity-building workshops in research methods, statistical analysis and scientific reporting.

PUBLICATIONS AND MANUSCRIPTS

Peer-Reviewed Publication

- **Nyabuto, E.**, Otto, P., & Okhrin, Y. (2026). Estimation of spatial and temporal autoregressive effects using LASSO: An application to hourly particulate matter concentrations in Bavaria, Germany. *Environmetrics*. https://doi.org/10.1002/env.70105

Manuscripts

- **Nyabuto, E.**, & Otto, P. Learning Volatility Dependence Networks in UK Equity Markets using Penalised Spatiotemporal Autoregressive Conditional Heteroscedasticity Models. *Under review*.
- **Nyabuto, E.** A Regularised Endemic–Epidemic Approach to Learning Epidemiological Connectivity from Disease Surveillance Data: Evidence from Malaria in Mozambique. *Manuscript in preparation*.

SELECTED RESEARCH PROJECTS AND COLLABORATIONS

University of Exeter, United Kingdom *July 2026*
European Study Group with Industry 195 (ESGI 195) – NEOM Nature Reserve Challenge

- Collaborated in an interdisciplinary team on statistical methods for estimating and forecasting vegetation carbon capture using Sentinel-2 satellite imagery, drone data, and environmental indicators.
- Contributed expertise in spatial modelling, data fusion, statistical calibration, and uncertainty quantification to translate multi-resolution remote-sensing data into vegetation carbon estimates. The project culminated in a technical research report for the NEOM Nature Reserve challenge.

AIMS Research and Innovation Centre, Kigali, Rwanda *August 2026*
Research Visit

- Conducted collaborative research on data-driven spatial dependence learning for epidemiological

surveillance and malaria risk modelling.
- Developed and evaluated geographic and learned dependence structures within an endemic–epidemic framework. The visit supported development of an ongoing manuscript on data-driven epidemiological dependence learning from malaria surveillance data.

SELECTED RESEARCH PRESENTATIONS AND CONFERENCES

- **Oral presentation:** "Learning Volatility Dependence Networks in UK Equity Markets using Penalised Spatiotemporal Autoregressive Conditional Heteroscedasticity Models.", Doctoral Training Programme in Data Science 2026, AIMS RIC, Rwanda Mar 2026
- **Oral presentation:** "Estimation of Spatial and Temporal Effects Using Lasso, An example of Particulate Matter Concentrations in Bavaria, Germany", Spatial Econometrics Conference (SEA 2024), University of Groningen, Netherlands Jun 2024
- **Oral presentation:** "Estimation of Spatial and Temporal Effects Using Lasso", Young Statisticians Workshop, Dortmund, Germany Sep 2023
- German Statistical Week 2023, Dortmund, Germany Sep 2023
- 18th Doktorandentreffen Stochastik, Heidelberg University, Germany Aug 2023

TEACHING AND MENTORING EXPERIENCE

University of Glasgow, Glasgow, UK *Jan 2024 – Present*
Teaching Assistant (Part-time), School of Mathematics and Statistics

- Deliver tutorials and laboratory sessions in probability, statistical inference, regression modelling, experimental design, data analysis, and quantitative methods.
- Support students in statistical computing, mathematical reasoning, model interpretation, and data analysis.
- Assess student work and provide methodological, statistical-computing and quantitative mentoring.

Kibabii University, Bungoma, Kenya *Jan 2021 – Aug 2022*
Tutorial Fellow (Part-time), Department of Mathematics

- Taught undergraduate Probability, Statistics and Estimation Theory, including assessment and mentoring in statistical computing and quantitative research.

PROFESSIONAL SERVICE, LEADERSHIP & MEMBERSHIPS

- Strategic Advisory Board Member, CoSE, University of Glasgow (2023 – Present)
- Student Member, Royal Statistical Society (2023 – Present)

ADDITIONAL PROFESSIONAL EXPERIENCE

Computer Science Academy, CSA Africa, Glasgow, UK *Feb 2025 – Aug 2025*
Funding & Partnerships Officer (Part-time)

- Coordinated partnerships and funding activities supporting the CSA Africa Python Workshop, which trained more than 200 early-career researchers and students across Africa.
- Supported research capacity-building and technical training initiatives in Python and data analytics and developed systems for coordinating funding opportunities and stakeholder engagement.

Crossroads International–NAFGEM Tanzania, Moshi, Tanzania *Nov 2021 – Aug 2022*
IT Specialist Advisor

- Designed and implemented a digital data-collection and management system that reduced operational costs by approximately 30% while strengthening programme monitoring, reporting, and evidence-based decision-making.
- Led IT and digital-literacy capacity building for staff and youth, improving the use of digital tools for data management, analysis, communication, and programme delivery.
- Applied data and technology solutions to support monitoring, evaluation, fundraising, and community programmes addressing FGM, child marriage, and youth empowerment.